example_ipbus

Version: 1.0

Wednesday 26th November, 2025 16:57

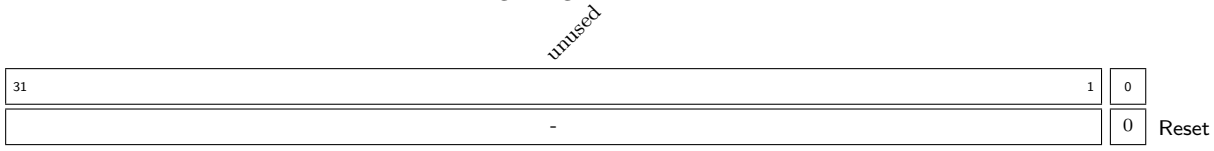
An example module that contain all the register types that are currently supported by bust.

1 Register List

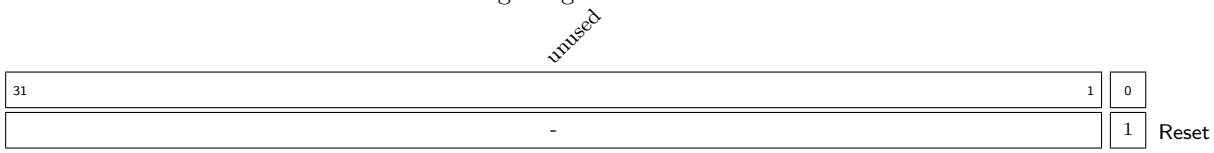
#	Name	Mode	Address	Type	Width	Reset
0	reg0	RW	0x00000000	BIT	1	0x0
1	reg1	RW	0x00000004	BIT	1	0x1
2	reg2	RO	0x00000008	BIT	1	0x0
3	reg3	RW/STALL	0x0000000C	VECTOR	8	0x3
4	reg4	RO	0x00000010	VECTOR	14	0x0
5	reg5	RW	0x00000014	DEFAULT	32	0xFFFFFFFF
6	reg6	RO	0x00000018	DEFAULT	32	0x0
7	reg7	RW	0x0000001C	FIELDS	21	0xAD7
8	reg8	RO	0x00000020	FIELDS	24	0x0
9	reg9	PULSE	0x00000024	BIT	1	0x1
10	reg10	PULSE	0x00000028	VECTOR	4	0xA
11	reg11	PULSE	0x0000002C	FIELDS	16	0x3

2 Registers

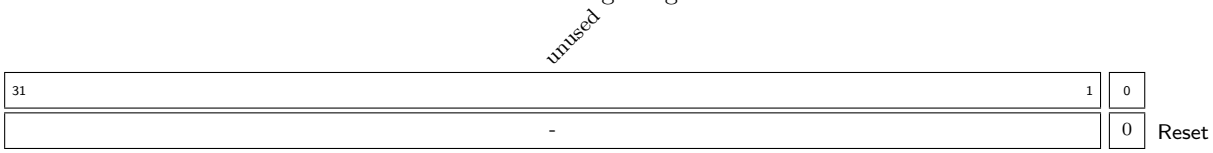
Register 2.1: REG0 - RW (0x00000000)
RW std_logic register that resets to 0x0



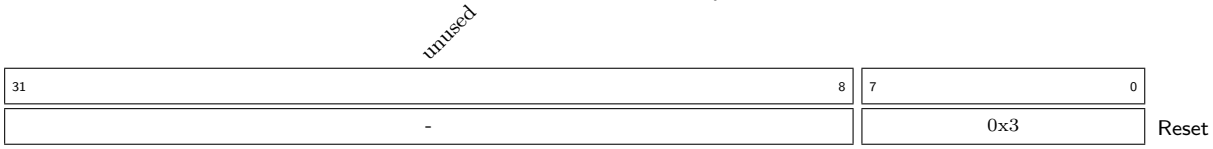
Register 2.2: REG1 - RW (0x00000004)
RW std_logic register that resets to 0x1



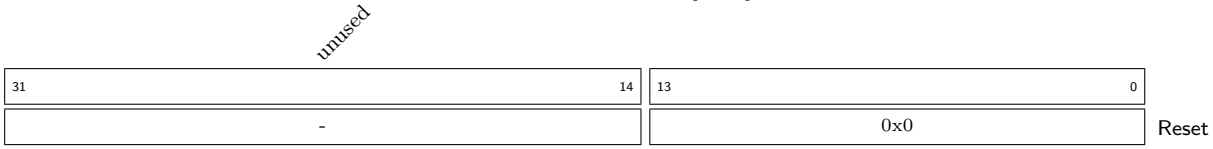
Register 2.3: REG2 - RO (0x00000008)
RO std_logic register



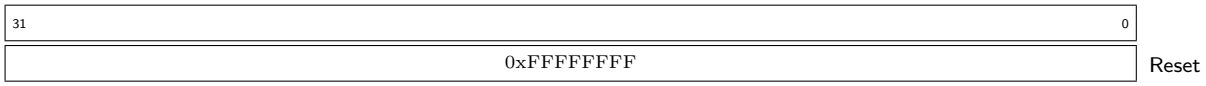
Register 2.4: REG3 - RW - STALL FOR 30 CYCLES (0x0000000C)
RW std_logic_vector[7:0] register that resets to 0x3. Reading and writing from this register is stalling the bus for 30 extra clock cycles.



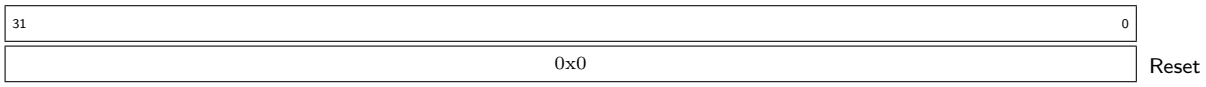
Register 2.5: REG4 - RO (0x00000010)
RO std_logic_vector[13:0]



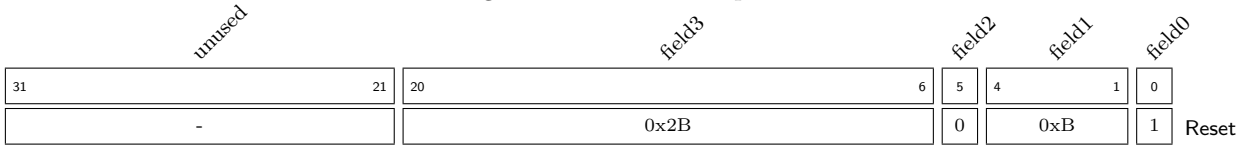
Register 2.6: REG5 - RW (0x00000014)
Default RW register that resets to 0xFFFFFFFF



Register 2.7: REG6 - RO (0x00000018)
Default RO register



Register 2.8: REG7 - RW (0x0000001C)
RW register that have multiple fields



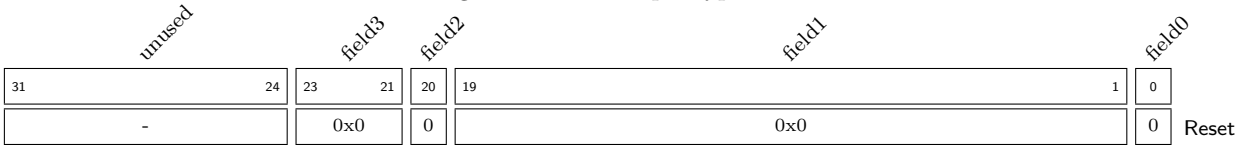
field0 std_logic that resets to 0x1

field1 std_logic_vector[3:0] that resets to 0xb is not a valid reset value

field2 std_logic that resets to 0x1

field3 std_logic_vector[14:0] that resets to 0x2b

Register 2.9: REG8 - RO (0x00000020)
RO register with multiple types of fields



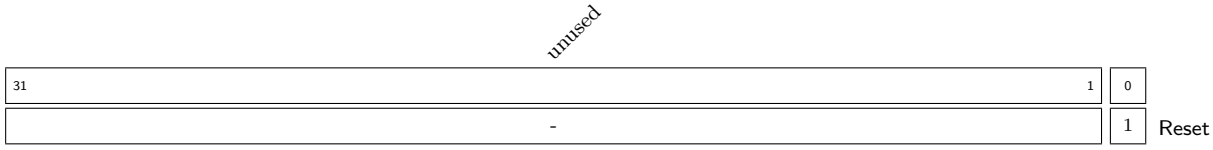
field0 std_logic field

field1 std_logic_vector[18:0] field

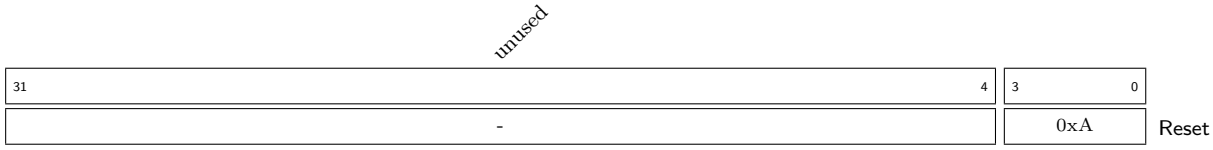
field2 std_logic field

field3 std_logic_vector[2:0] field

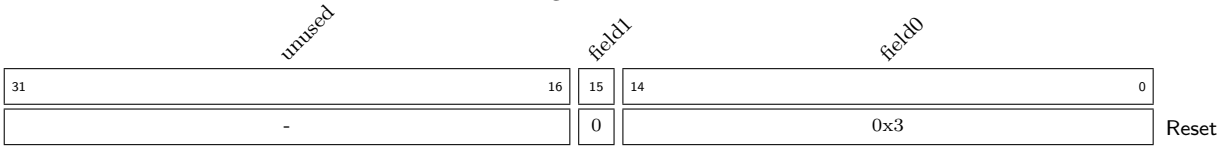
Register 2.10: REG9 - PULSE FOR 4 CYCLES (0x00000024)
PULSE std_logic register that resets to 0x1



Register 2.11: REG10 - PULSE FOR 1 CYCLES (0x00000028)
PULSE std_logic_vector[3:0] register that resets to 0xA



Register 2.12: REG11 - PULSE FOR 50 CYCLES (0x0000002C)
PULSE register with two fields



field0 std_logic_vector[14:0] field that resets to 0x3

field1 std_logic field that resets to 0x0